DIB2 interface, USB PHY

File: DIB_interface.kicad_sch

MCU, SDRAM

File: MCU.kicad_sch

ADIB, PWM, AFE connectors

File: ADIB_PWM.kicad_sch

Voltage_current DACs

File: Voltage_current_DACs.kicad_sch

Voltage DAC, IO expaders, Digital outputs

File: Voltage_DAC_Digital_outputs.kicad_sch

Digital inputs

File: Digital_inputs.kicad_sch





KK1 PCB-MB-01 KK2 PCB-MB-01

https://www.envox.eu

Mixed I/O module for BB3+

Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)

Envox d.o.o.

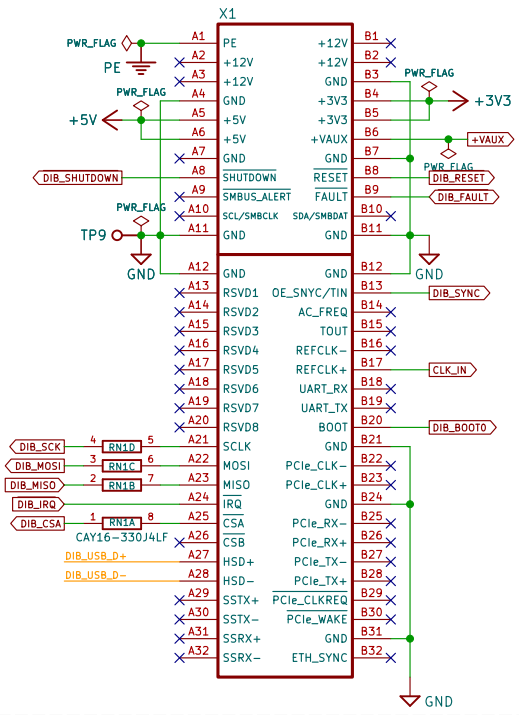
Sheet: /

File: EEZ DIB MIO168plus.kicad_sch

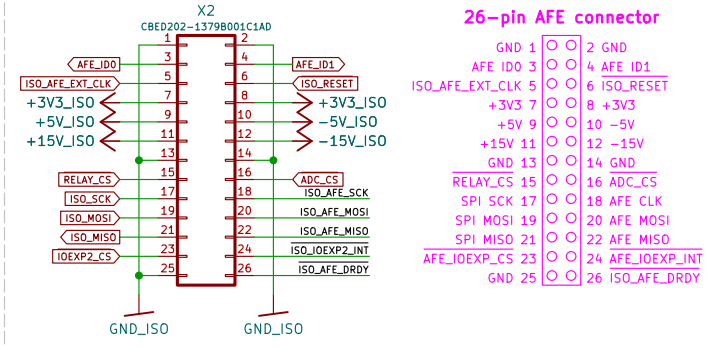
Title: EEZ DIB MIO168+ module

Size: A4	Date: 2026-10-08	**Rev: r1B2**
KiCad E.D.A. 10.0.6		Id: 1/7

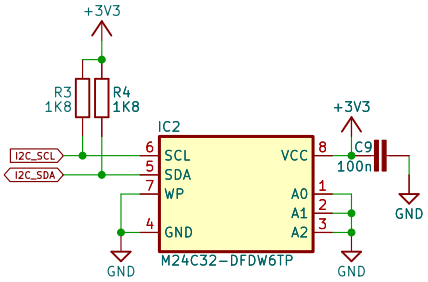
DIB2 connector



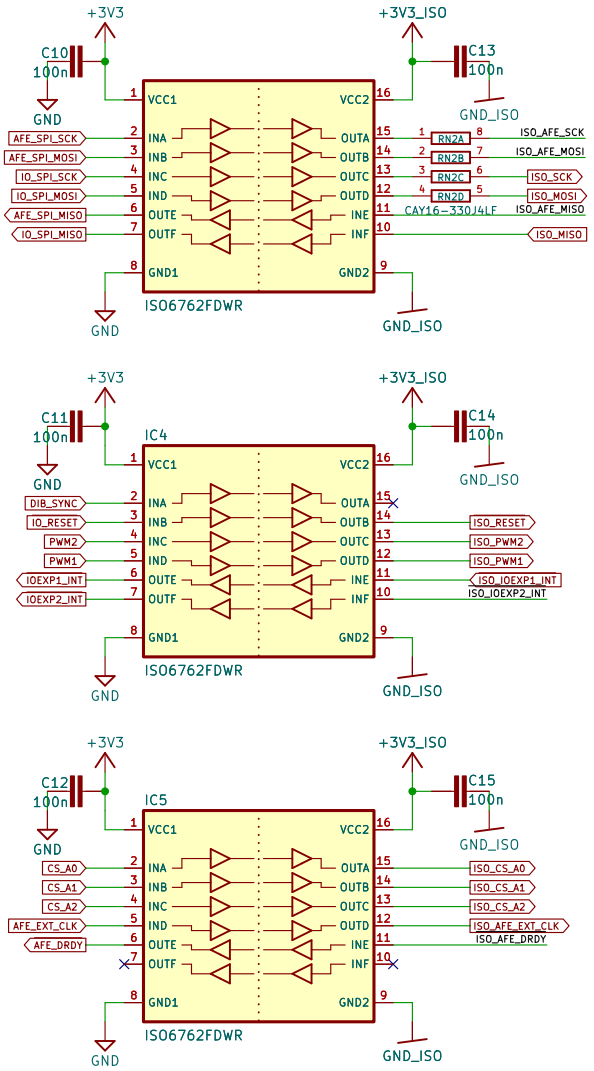
Isolated AFE connector



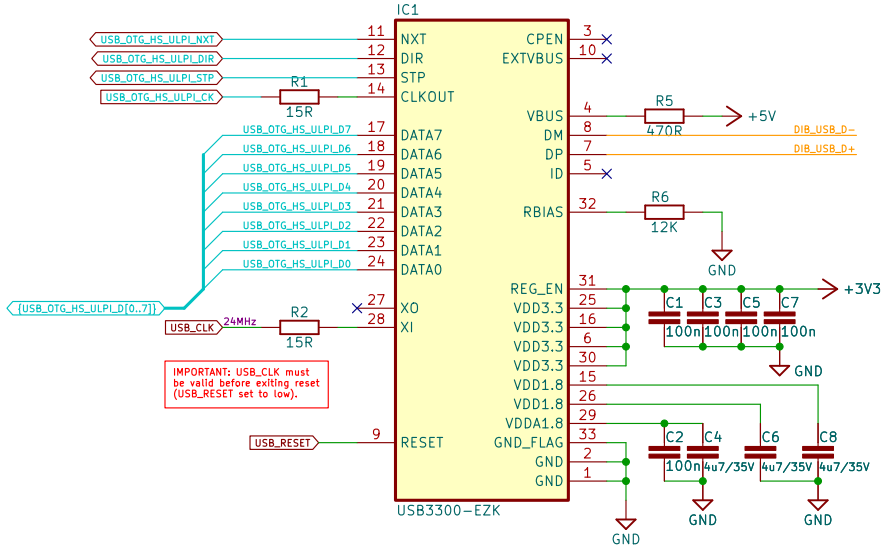
I2C Module ID EEPROM



Bus Isolator



USB HS PHY



https://www.envox.eu
Mixed I/O module for BB3+
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /DIB2 interface, USB PHY/
File: DIB_interface.kicad_sch

Title: EEZ DIB MIO168+ module

Size: A4	Date: 2026−10−08	**Rev: r1B2**
KiCad E.D.A. 10.0.6		Id: 2/7

32-bit MCU

Ensure that SPI is selected for the bootloader mode in accordance with the instructions in AN2606.

IC6A

STM32F446ZET6

IC6B

STM32F446ZET6

IC6C

STM32F446ZET6

SDRAM (64kbit, 4MBx16)

IMPORTANT: For PCB layout consult ST AN4661 section 8.4.2

IC7

IS42S16400N-6TL

Consumption: +3.3 V, up to 110 mA

PCB pads for WR-WST REDFIT IDC

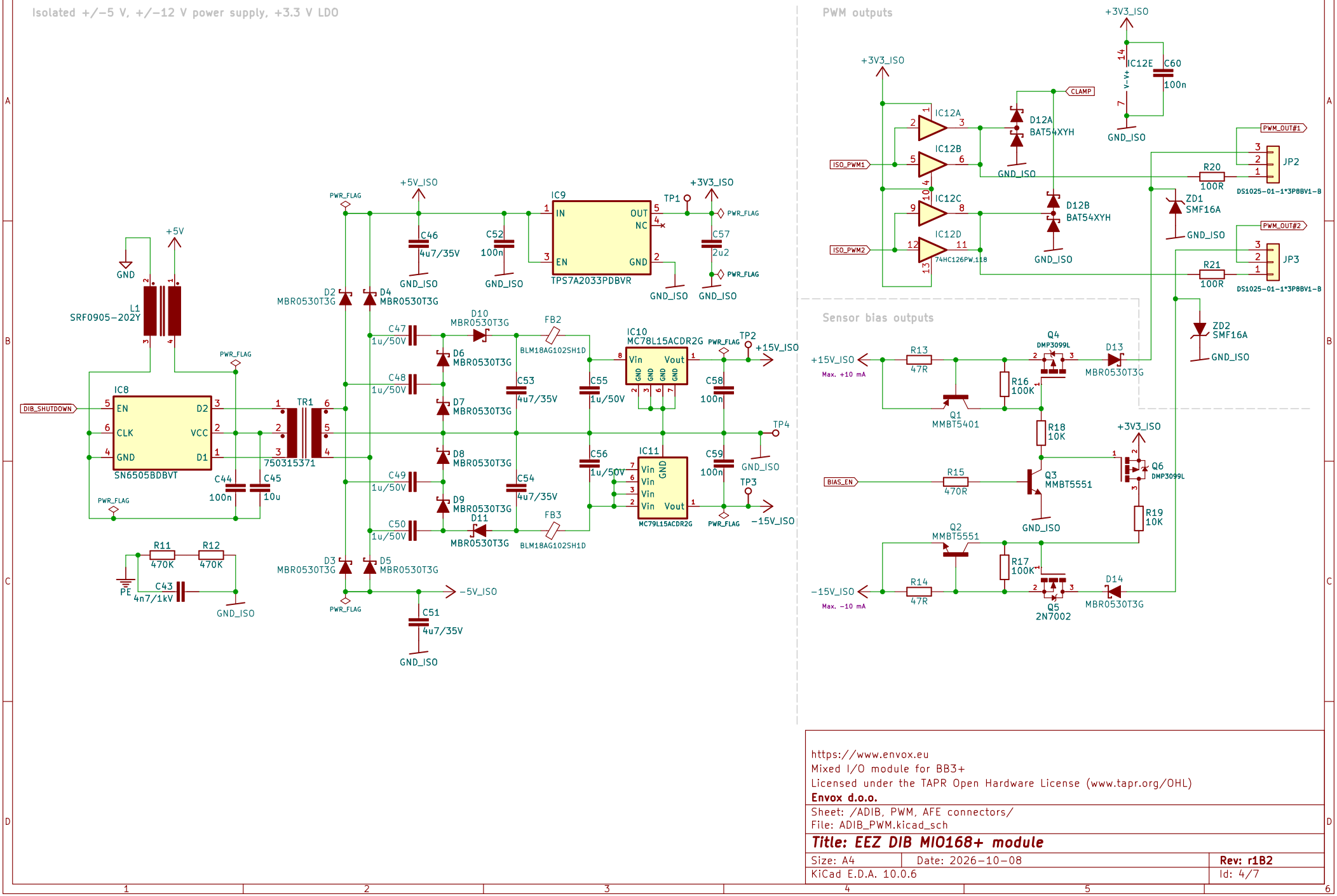
X3

https://www.envox.eu
Mixed I/O module for BB3+
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

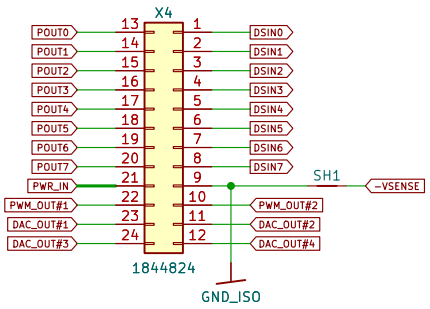
Sheet: /MCU, SDRAM/
File: MCU.kicad_sch

Title: EEZ DIB MIO168+ module

Size: A4	Date: 2026-10-08	**Rev: r1B2**
KiCad E.D.A. 10.0.6		Id: 3/7


Isolated +/−5 V, +/−12 V power supply, +3.3 V LDO
PWM outputs
Sensor bias outputs
+5V
GND
L1
SRF0905-202Y
PWR_FLAG
IC8
SN6505BDBVT
DIB_SHUTDOWN
TR1
750315371
C44
100n
C45
10u
R11
470K
R12
470K
C43
4n7/1kV
PE
GND_ISO
+5V_ISO
C46
4u7/35V
C52
100n
IC9
TPS7A2033PDBVR
TP1
+3V3_ISO
C57
2u2
D2
MBR0530T3G
D4
MBR0530T3G
D3
MBR0530T3G
D5
MBR0530T3G
D6
MBR0530T3G
D7
MBR0530T3G
D8
MBR0530T3G
D9
MBR0530T3G
D10
MBR0530T3G
D11
MBR0530T3G
C47
1u/50V
C48
1u/50V
C49
1u/50V
C50
1u/50V
C51
4u7/35V
C53
4u7/35V
C54
4u7/35V
FB2
BLM18AG102SH1D
FB3
BLM18AG102SH1D
IC10
MC78L15ACDR2G
IC11
MC79L15ACDR2G
C55
1u/50V
C56
1u/50V
C58
100n
C59
100n
TP2
+15V_ISO
TP3
−15V_ISO
TP4
−5V_ISO
IC12A
IC12B
IC12C
IC12D
74HC126PW,118
IC12E
C60
100n
ISO_PWM1
ISO_PWM2
CLAMP
D12A
BAT54XYH
D12B
BAT54XYH
R20
100R
R21
100R
PWM_OUT#1
PWM_OUT#2
JP2
JP3
DS1025-01-1*3P8BV1-B
ZD1
SMF16A
ZD2
SMF16A
R13
47R
R14
47R
R15
470R
R16
100K
R17
100K
R18
10K
R19
10K
Q1
MMBT5401
Q2
MMBT5551
Q3
MMBT5551
Q4
DMP3099L
Q5
2N7002
Q6
DMP3099L
D13
MBR0530T3G
D14
MBR0530T3G
BIAS_EN
Max. +10 mA
Max. −10 mA
https://www.envox.eu
Mixed I/O module for BB3+
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.
Sheet: /ADIB, PWM, AFE connectors/
File: ADIB_PWM.kicad_sch
Title: EEZ DIB MIO168+ module
Size: A4
Date: 2026-10-08
Rev: r1B2
KiCad E.D.A. 10.0.6
Id: 4/7

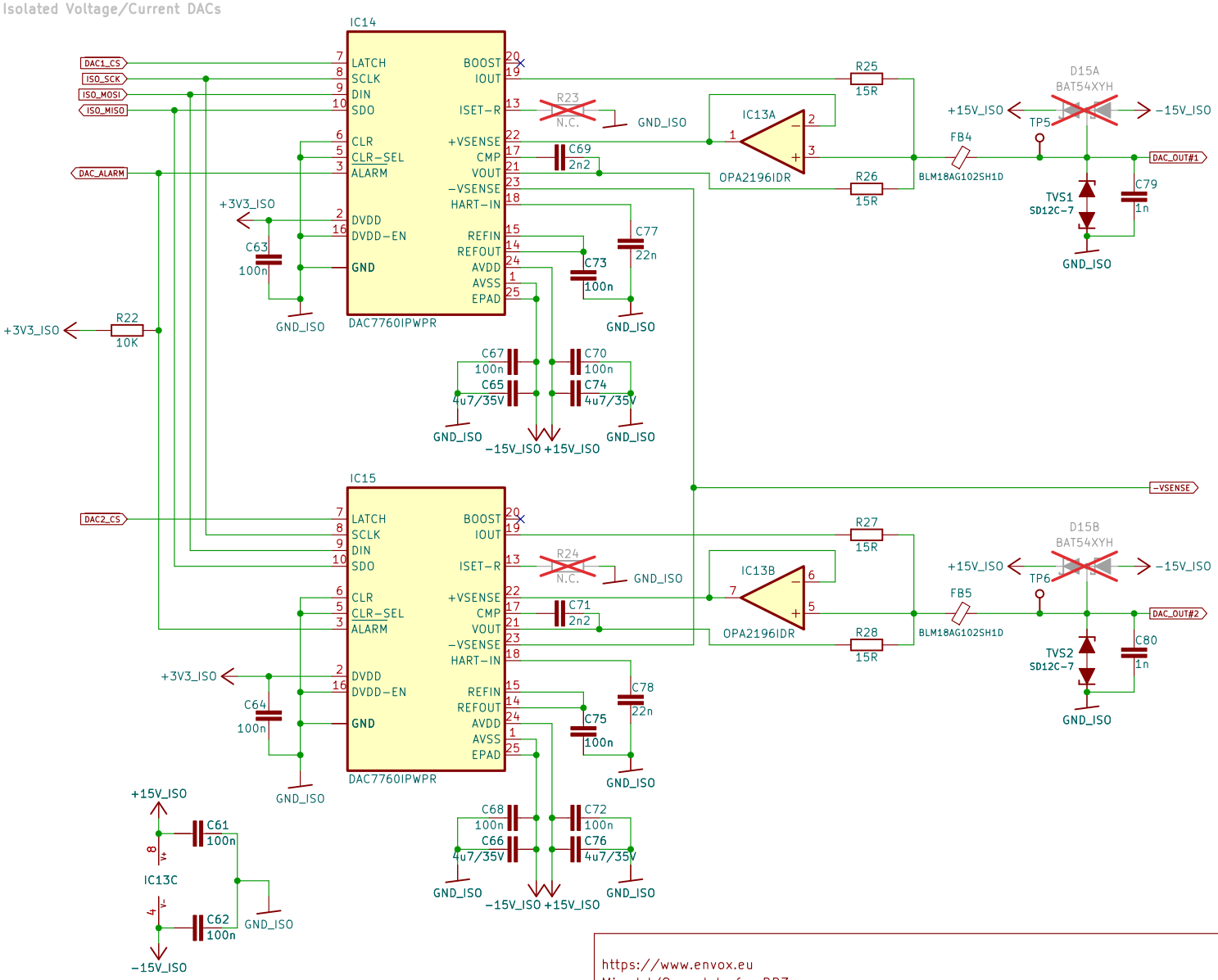
Isolated Front panel I/O connector



24-pin I/O connector



Isolated Voltage/Current DACs



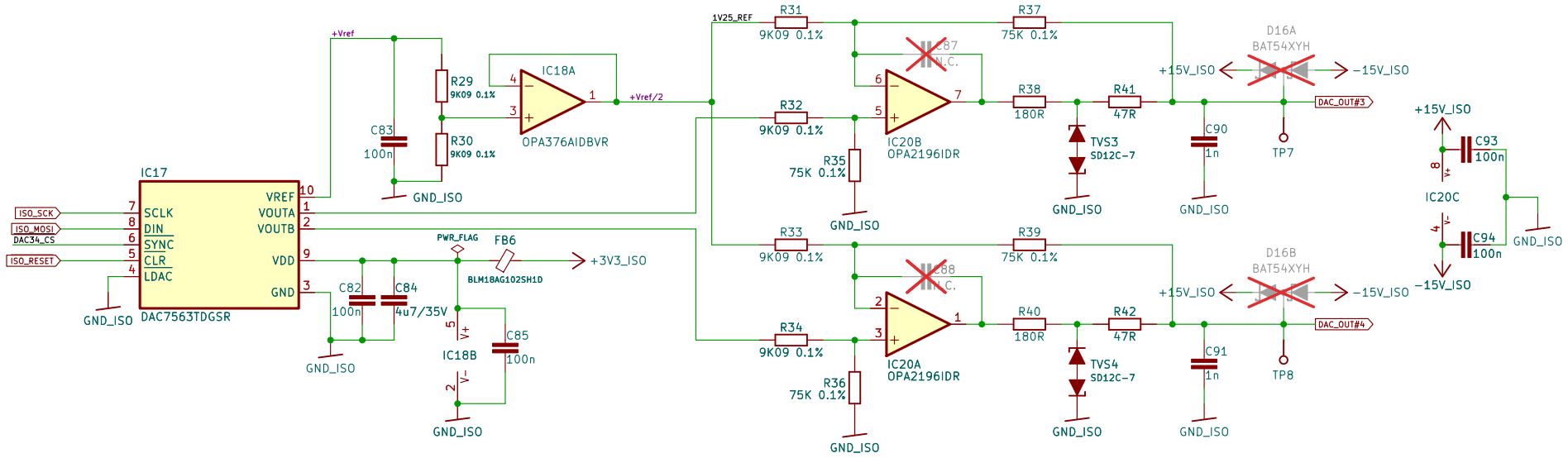
https://www.envox.eu
Mixed I/O module for BB3+
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /Voltage_current DACs/
File: Voltage_current_DACs.kicad_sch

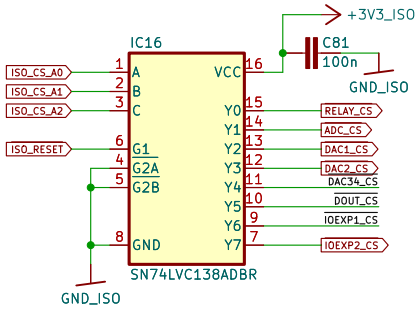
Title: EEZ DIB MIO168+ module

Size: A4	Date: 2026-10-08	**Rev: r1B2**
KiCad E.D.A. 10.0.6		Id: 5/7

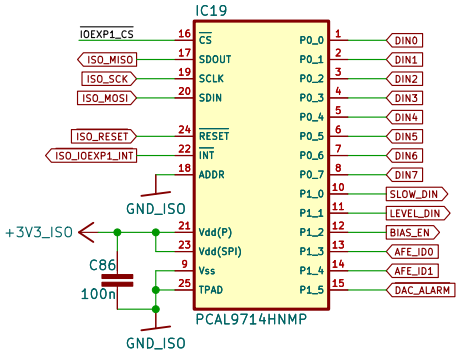
Isolated 2-ch DAC with bipolar output



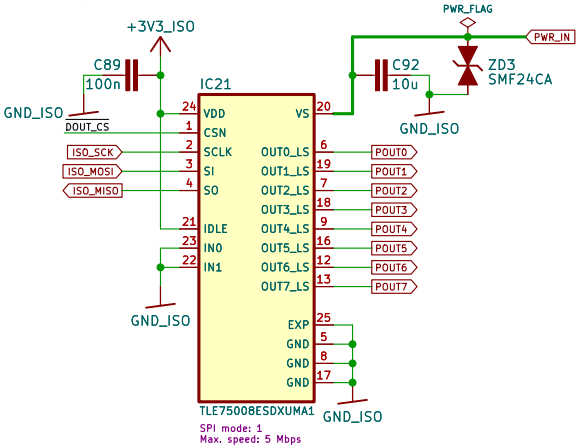
Isolated SPI CS decoder



Isolated 14-bit I/O Expander



Isolated 8-ch digital power outputs



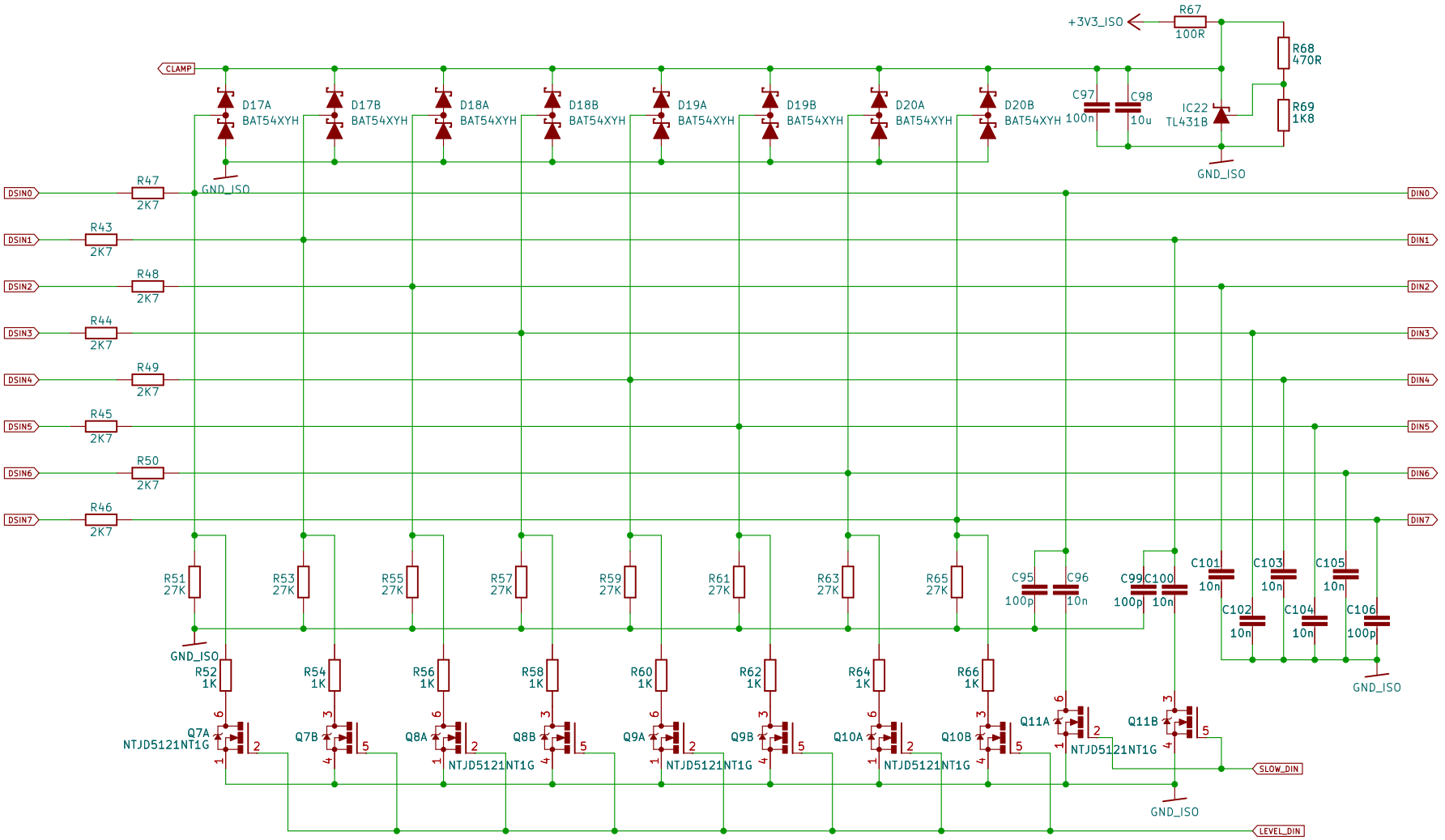
https://www.envox.eu
Mixed I/O module for BB3+
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /Voltage DAC, IO expaders, Digital outputs/
File: Voltage_DAC_Digital_outputs.kicad_sch

Title: EEZ DIB MIO168+ module

Size: A4	Date: 2026-10-08	**Rev: r1B2**
KiCad E.D.A. 10.0.6		Id: 6/7

Digital inputs



https://www.envox.eu
Mixed I/O module for BB3+
Licensed under the TAPR Open Hardware License (www.tapr.org/OHL)
Envox d.o.o.

Sheet: /Digital inputs/
File: Digital_inputs.kicad_sch

Title: EEZ DIB MIO168+ module

Size: A4	Date: 2026-10-08	**Rev: r1B2**
KiCad E.D.A. 10.0.6		Id: 7/7